

Q1

1. \$1000 USD

Compatibility: See details below.							Estimated Wattage: 237W		
Component	Selection	Base	Promo	Shipping	Tax	Availability	Price	Where	
CPU	 Intel Core i5-14400 2.5 GHz 10-Core Processor	\$191.99	—	—	—	In stock	 \$191.99	amazon.com	Buy ×
CPU Cooler	+ Choose A CPU Cooler								
Motherboard	 MSI PRO Z790-A MAX WIFI ATX LGA1700 Motherboard	\$224.99	—		—	In stock	 \$224.99	amazon.com	Buy ×
Memory	 Crucial Classic 16 GB (2 x 8 GB) DDR5-5600 CL46 Memory	\$54.99	—		—	Available soon	 \$54.99	amazon.com	Buy ×
	+ Add Additional Memory								
Storage	 Samsung 990 Pro 2 TB M.2-2280 PCIe 4.0 X4 NVME Solid State Drive	\$159.99	—	FREE	—	In stock	 \$159.99	Abt	Buy ×
	+ Add Additional Storage								
Video Card	+ Choose A Video Card								
Case	 Phanteks XT PRO ATX Mid Tower Case	\$69.99	-\$10.00 ¹	\$8.99	—	In stock	 \$68.98	newegg	Buy ×
Power Supply	 MSI MAG A550BN 550 W 80+ Bronze Certified ATX Power Supply	\$49.99	—		—	In stock	 \$49.99	amazon.com	Buy ×
Operating System	 Microsoft Windows 11 Home OEM - DVD 64-bit	\$119.99	-\$10.00 ²	FREE	—	In stock	 \$109.99	newegg	Buy ×
Monitor	+ Choose A Monitor								
Expansion Cards / Networking	Sound Cards, Wired Network Adapters, Wireless Network Adapters								
Peripherals	Headphones, Keyboards, Mice, Speakers, Webcams								
Accessories / Other	Case Accessories, Case Fans, Fan Controllers, Thermal Compound, External Storage, Optical Drives, UPS Systems								
							Base Total:	\$871.93	
							Promo Discounts:	-\$10.00	
							Mail-in Rebates:	-\$10.00	
							Shipping:	\$8.99	
							Total:	\$860.92	

2. \$2000 USD

Compatibility: See details below.							Estimated Wattage: 454W		
Component	Selection	Base	Promo	Shipping	Tax	Availability	Price	Where	
CPU	 Intel Core i9-10980XE 3 GHz 18-Core Processor	\$849.00	—	—	—	In stock	\$849.00	amazon.com	Buy X
CPU Cooler	 Cooler Master Hyper 212 Black Edition 42 CFM CPU Cooler	\$25.99	—	Prime	—	In stock	\$25.99	amazon.com	Buy X
Motherboard	 Asus Prime X299-A II ATX LGA2066 Motherboard	\$245.99	—	Prime	—	In stock	\$245.99	amazon.com	Buy X
Memory	 Corsair Vengeance LPX 32 GB (2 x 16 GB) DDR4-3200 CL16 Memory	\$82.99	-\$3.00 ¹	FREE	—	In stock	\$79.99	newegg	Buy X
	+ Add Additional Memory								
Storage	 Samsung 990 Pro 2 TB M.2-2280 PCIe 4.0 X4 NVME Solid State Drive	\$159.99	—	FREE	—	In stock	\$159.99	Abt	Buy X
	+ Add Additional Storage								
Video Card	 MSI GeForce RTX 3060 Ventus 2X 12G GeForce RTX 3060 12GB 12 GB Video Card	\$293.85	—	Prime	—	In stock	\$293.85	amazon.com	Buy X
	+ Add Another Video Card								
Case	 Montech XR ATX Mid Tower Case	\$73.73	—	Prime	—	In stock	\$73.73	amazon.com	Buy X
Power Supply	 MSI MAG A750GL PCIES 750 W 80+ Gold Certified Fully Modular ATX Power Supply	\$99.99	—	Prime	—	In stock	\$99.99	amazon.com	Buy X
Operating System	 Microsoft Windows 11 Home OEM - DVD 64-bit	\$119.99	-\$10.00 ²	FREE	—	In stock	\$109.99	newegg	Buy X
Monitor	+ Choose A Monitor								
Expansion Cards / Networking	Sound Cards, Wired Network Adapters, Wireless Network Adapters								
Peripherals	Headphones, Keyboards, Mice, Speakers, Webcams								
Accessories / Other	Case Accessories, Case Fans, Fan Controllers, Thermal Compound, External Storage, Optical Drives, UPS Systems								
							Base Total:	\$1951.52	
							Promo Discounts:	-\$13.00	
							Total:	\$1938.52	

3. \$4000 USD

Compatibility: See details below.								⚡ Estimated Wattage: 776W		
Component	Selection		Base	Promo	Shipping	Tax	Availability	Price	Where	
CPU		Intel Xeon E5-2699 V4 2.2 GHz 22-Core OEM/Tray Processor	\$349.00	—	—	—	In stock	⚙️ \$349.00	amazon.com	Buy ✕
CPU		Intel Xeon E5-2699 V4 2.2 GHz 22-Core OEM/Tray Processor	\$349.00	—	—	—	In stock	⚙️ \$349.00	amazon.com	Buy ✕
CPU Cooler		Cooler Master Hyper 212 Black Edition 42 CFM CPU Cooler	\$25.99	—	📦 Prime	—	In stock	⚙️ \$25.99	amazon.com	Buy ✕
CPU Cooler		Cooler Master Hyper 212 Black Edition 42 CFM CPU Cooler	\$25.99	—	📦 Prime	—	In stock	⚙️ \$25.99	amazon.com	Buy ✕
Motherboard		Supermicro MBD-X10DAX EATX Dual-CPU LGA2011-3 Motherboard	\$900.00	—	—	—	In stock	⚙️ \$900.00	amazon.com	Buy ✕
Memory		Kingston Server Premier 64 GB (1 x 64 GB) Registered DDR4-3200 CL22 Memory	\$299.00	—	—	—	In stock	⚙️ \$299.00	amazon.com	Buy ✕
	+ Add Additional Memory									
Storage		Seagate Barracuda Compute 2 TB 3.5" 7200 RPM Internal Hard Drive	\$64.98	—	—	—	In stock	⚙️ \$64.98	amazon.com	Buy ✕
		Seagate Barracuda Compute 2 TB 3.5" 7200 RPM Internal Hard Drive	\$64.98	—	—	—	In stock	⚙️ \$64.98	amazon.com	Buy ✕
	+ Add Additional Storage									
Video Card		NVIDIA TITAN RTX TITAN RTX 24 GB Video Card	\$1134.73	—	—	—	In stock	⚙️ \$1134.73	amazon.com	Buy ✕
	+ Add Another Video Card									
Case		Phanteks XT PRO ATX Mid Tower Case	\$69.99	-\$10.00 ¹	\$8.99	—	In stock	⚙️ \$68.98	newegg.com	Buy ✕
Power Supply		MSI MAG A850GL PCIES 850 W 80+ Gold Certified Fully Modular ATX Power Supply	\$119.99	—	📦 Prime	—	In stock	⚙️ \$119.99	amazon.com	Buy ✕
Operating System		Microsoft Windows 8.1 Pro OEM 64-bit	\$179.00	—	\$4.01	—	In stock	⚙️ \$183.01	amazon.com	Buy ✕
Monitor	+ Choose A Monitor									
Expansion Cards / Networking	Sound Cards, Wired Network Adapters, Wireless Network Adapters									
Peripherals	Headphones, Keyboards, Mice, Speakers, Webcams									
Accessories / Other	Case Accessories, Case Fans, Fan Controllers, Thermal Compound, External Storage, Optical Drives, UPS Systems									
								Base Total:	\$3582.65	
								Mail-in Rebates:	-\$10.00	
								Shipping:	\$13.00	
								Total:	\$3585.65	
Buy From Amazon										

Q2.

品 名	產 品 名 稱	備 註	數 量	小 計
主機板 MB 全頁	【狂】華碩 PRIME B660M-A WIFI D4-CSM + Intel i7-12700【12 核/20 緒】，\$14690 ◆ ★	Intel B660 / 1700 腳位 (DDR4)-12 代.13 代皆支援	1	14690
記憶體 RAM 全頁	威剛 32GB DDR4-3200(AD4U3200732G22-SGN)(2048*8), \$2699 ◆ ★ 熱賣	桌上型記憶體 DDR4	1	2699
固態硬碟 M.2 SSD	威剛 XPG S70BLADE 1T/Gen4 讀:7400M/寫:5500M/ DRAM 快取/五年/贈散熱片 ▼下殺到 10/11 10:00, \$3230  \$3099 ◆ ★	M.2 PCIe 4.0 (Gen4) SSD 固態硬碟	1	3099
顯示卡 VGA 點我 全頁	華碩 DUAL-GTX1650-O4G(1725MHz/20.4cm/雙風扇/註四年)*開學活動優惠(10/31 止), \$4990 ◆ ★ 熱賣 ↓任搭 250↓	NVIDIA GTX1650	1	4990
CASE 機殼(+ 電源) 全頁	華碩 Prime AP201 黑/方形進氣孔/前 Type-C/M-ATX+華碩 TUF GAMING 550W 銅牌電源, \$3490 ◆ ★ 熱賣	華碩 ASUS 機殼	1	3490
燒錄器 CD/DVD/BD	LiteOn EB1 輕薄外接式 DVD 藍光燒錄器/支援 Mac, \$2680 ◆ ★	USB 外接藍光燒錄器	1	2680
TDP 耗電 280 瓦 2022/10/08 22:39:20 星期六 含稅 現金價：31648 優惠省 250 現金優惠價：31398				

Q3.

1. Dual-core, quad-core, hex-core, and oct-core.

多核心（雙核、四核、六核、八核）能增加多工運算的執行速度，增加可以獨立執行指令的運算資源。

2. DDR (Double data rate), dual channel, and quad channel

DDR 是記憶體傳輸技術，匯流排會在定時器訊號的上、下邊緣傳輸資料，傳輸速度是早期 SDR(Single data rate，在一次訊號傳輸一次)的兩倍。

Dual Channel 代表主機板一次能夠存取兩通道的記憶體，使得頻寬可以達到兩倍(Dual)。

Quad Channel 則達到四倍

3. PCI-e, USB, and thunderbolt

三者都是資料傳輸的介面，PCI-e 常見於電腦主機板的線路，USB 為近年行動裝置或連接非常常見的介面，而 Thunderbolt 則是近幾年出現能夠承載高速、高供電、高頻寬的介面。

4. SATA, Hybrid drives, and SAS (Serial Attached SCSI).

SATA 是常見的儲存裝置相關傳輸介面，常見於主機板連接硬碟、SSD、光碟機等

Hybrid drive 是結合 HDD 和 SSD，有傳統硬碟的磁盤、但也有內建快閃記憶體，讓 Hybrid Drive 的存取速度接近 SSD。

SAS 是一種電腦周邊零件傳輸的集線技術，常與 SATA 相容，連接硬碟、SSD、光碟機等。

5. SAN, NAS, and DAS storages.

常見的儲存架構，DAS 是直接用纜線連接到裝置上，如外接硬碟。

NAS 是透過乙太網路提供檔案儲存空間的設備，有許多協定方便共享。

SAN 是專用於連接伺服器與網路的高速網路，形成一個集中式的 Storage Pool.

6. RS-232, USB, and Thunderbolt interfaces.

三者都是資料傳輸的介面，RS-232 為序列資料的通訊標準，USB 為近年行動裝置或連接非常常見的介面，而 Thunderbolt 則是近幾年出現能夠承載高速、高供電、高頻寬的介面。

7. RJ-45.

為乙太網路用的接頭，常見於路由器、交換器與連網裝置(電腦)的连接。

8. Integrated (Onboard) VGA and discrete VGA.

內建顯示晶片與獨立顯示卡，Integrated VGA 常內建在 CPU 上，成本低，但效能僅滿足文書使用，如果要多 GPU 加速、3D 遊戲、特效等，就需要 Discrete VGA 另外運算。Discrete VGA 成本、散熱、耗電通常較高，配有 VRAM 而不需與系統 RAM 搶資源。